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CMPG 325 NETWORKING PROJECT

TECHNICAL REPORT

BACKGROUND

The Client requested a small network for a Health Care Centre, the proposed network plan was initially laid out requiring an ipv6 dual stack network plan in an extended star topology.

CHALLENGES

1. The physical topology which was proposed in the initial plan required a lot of switches. The configuration and connecting for each switch was challenging and posed a concern on the functioning of the network.
2. The proposed network challenge as requested by the client namely dual stack, initially gave the service provider a challenge especially IP addressing translation from ipv4 to ipv6
3. Configuration of the Integrated Services Router(IRS) in order to implement ipv6 was challenging initially.
4. Meshing up the logical design with the physical outcome of the network was challenging, some amendments had to be applied changing the initial plan as proposed in the beginning of the project.
5. Budget adjustments had to be done to ensure the final proposed network is completed. Extra costs were then catered for.

IMPLEMENTATION DETAILS

To successfully deploy the IPv6 dual-stack infrastructure alongside IPv4 for the healthcare center's Electronic Health Records (EHR) and administrative network, the following technical configurations were implemented:

- **Global Routing Activation:** Enabled IPv6 unicast routing globally across both the core router and the Layer 3 switch using ipv6 unicast-routing to allow forwarding of IPv6 packets.
- Addressing Scheme & Subnetting: Allocated the IPv6 global unicast subnet 2001:db8:10:11:10::/64 across the core VLAN 10 domain.
- Gateway Configuration:
    - Configured the edge router interface (GigabitEthernet0/0) with 2001:db8:10:11:10::1/64.
    - Assigned the core Layer 3 switch SVI (VLAN 10) the gateway address 2001:db8:10:11:10::ff/64.
- Static Routing & Default Gateways: Established static default routes (ipv6 route ::/0 2001:db8:10:11:10::1) on the Layer 3 switch and downstream access switches to forward external traffic directly to the router.

- End-Device and Management Integration: Assigned static IPv6 addresses (starting from 2001:db8:10:11:10::2 onwards) to PCs and switch management interfaces, ensuring seamless dual-stack operation.

RESULTS

- Dual-Stack Connectivity: Successfully achieved simultaneous, reliable transmission of both IPv4 and IPv6 traffic across all segments of the extended star topology.
- Trunk and Native VLAN Stability: Resolved initial CDP native VLAN mismatch warnings by standardizing native VLAN 10 across all inter-switch trunk links, ensuring clean 802.1Q tagging.
- Seamless Inter-VLAN and External Routing: End-user PCs on downstream access switches successfully established end-to-end connectivity with the core Layer 3 switch and edge router, verified via successful ping tests for both protocol versions.
- Optimized Physical Footprint: By replacing excessive downstream branching with a centralized Layer 3 switch feeding four structured access switches, the physical wiring complexity was significantly reduced.

CONCLUSION AND RECOMMENDATIONS

Conclusion: The dual-stack network implementation for the healthcare center successfully modernizes the IT infrastructure, ensuring robust support for critical Electronic Health Records (EHR), patient check-in, and administrative workflows. Overcoming initial topology, IPv6 translation, and routing hurdles resulted in a resilient, scalable network architecture capable of meeting current operational demands while maintaining high performance.

FUTURE RECOMMENDATIONS

1. Dynamic IP Assignment: Transition from static IPv6 assignments to DHCPv6 (Stateful/Stateless SLAAC) to simplify device onboarding and reduce administrative overhead as the facility expands.
2. Dynamic Routing Protocols: Implement OSPFv3 as the network scales to replace static default routes, enabling automatic path recalculation and fault tolerance.
3. High Availability: Introduce redundant uplinks and Spanning Tree Protocol enhancements (such as Rapid PVST+ or MSTP) between the core Layer 3 switch and access switches to eliminate single points of failure.